

Why the 2-Selmer ranks agree

$y^2 = x^3 + a$ and $y^2 = x^3 - a^{-1}$ share their mod-2 Galois module; what that explains, and what it leaves to check

on a question of Rene Pannekoek, MathOverflow 227987 (2015)

1. The question

Numerically, the Mordell–Weil ranks of $y^2 = x^3 + p^3$ and $y^2 = x^3 - p^3$ almost always agree, the first exceptions being $p = 37, 61, 157, 193$. The curves are **not isogenous**, so the correlation is puzzling. Looking instead at 2-Selmer ranks makes the pattern exact: they appear to agree for **every** prime $p > 2$ — including at $p = 37$, where the ranks are 0 and 2 but both 2-Selmer ranks are 3. The general form conjectured is:

Conjecture. For a an **odd** integer, the elliptic curves

$$E_a : y^2 = x^3 + a \quad \text{and} \quad E_{-1/a} : y^2 = x^3 - a^{-1} \cong y^2 = x^3 - a^5$$

have equal 2-Selmer ranks.

The original statement is the case $a = p^3$; the variants with $x^3 \pm p$ against $x^3 \mp p^5$, and $x^3 \pm p^2$ against $x^3 \mp p^4$, are $a = \pm p$ and $a = \pm p^2$.

Note that $a \mapsto -1/a$ is an **involution**, and that it is the natural one: the curves $y^2 = x^3 + a$ are the sextic twists of one another, the twisting parameter lives in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^6$, and $-1/a \equiv -a^5$ there.

2. The answer to “why are they correlated”

The 2-Selmer group of E is cut out of $H^1(\mathbb{Q}, E[2])$ by local conditions. So two curves with **isomorphic mod-2 Galois modules** have their 2-Selmer groups inside the **same** group, and their ranks are constrained to be close whether or not the curves are isogenous. That is exactly what happens here, and it is visible in one line.

Theorem A. Let $a \in \mathbb{Q}^\times$ and let $A_b = \mathbb{Q}[t]/(t^3 + b)$. Then

$$\theta \mapsto -1/\theta$$

is an isomorphism of $\mathbb{Q}$ -algebras $A_a \rightarrow A_{-1/a}$. Consequently

$$E_a[2] \cong E_{-1/a}[2]$$

as $G_{\mathbb{Q}}$ -modules, and the two 2-Selmer groups are subgroups of one and the same $H^1(\mathbb{Q}, M)$.

Proof. If $\theta^3 = -a$ then $(-1/\theta)^3 = -1/\theta^3 = 1/a$, so $-1/\theta$ is a root of $t^3 - 1/a$; the map is a $\mathbb{Q}$ -algebra map between 3-dimensional algebras sending a generator to a generator, hence an isomorphism. For an elliptic curve $y^2 = f(x)$ with f cubic, $E[2]$ is the $G_{\mathbb{Q}}$ -set of roots of f — the non-zero points are $(\theta_i, 0)$ — with its induced $\mathbb{F}_2$ -structure. An isomorphism of the étale algebras is an isomorphism of the root sets as $G_{\mathbb{Q}}$ -sets, hence of the modules. ■

So the curves, though non-isogenous, have the same 2-division field $\mathbb{Q}(\sqrt[3]{a}, \zeta_3)$ and the same mod-2 representation. Checked numerically on $a = 3, 5, 7, 9, 11, 13, 15, 25, 27, -5, -7, 49, 121$: the factorisation types of $x^3 + a$ and $x^3 - a^5$ agree at every good prime up to 300, and the 2-division fields are isomorphic.

This already answers the question as asked. “Why would the ranks of non-isogenous curves be correlated?” — because rank correlation is not what is happening. The **2-Selmer** ranks are correlated because the two curves share the object that 2-Selmer ranks are computed from. The Mordell–Weil

ranks then agree whenever $\text{III}[2]$ does not intervene, and $p = 37$ is precisely a case where it does: the ranks are 0 and 2, the 2-Selmer ranks both 3, the difference being $\dim \text{III}[2] = 2$ on one side and 0 on the other.

3. What remains: a purely local statement

Fix $M = E_a[2] = E_{-1/a}[2]$ and write $L_v(E) \subseteq H^1(\mathbb{Q}_v, M)$ for the image of $E(\mathbb{Q}_v)/2$. Both $L_v(E_a)$ and $L_v(E_{-1/a})$ are **maximal isotropic** subspaces of the same $H^1(\mathbb{Q}_v, M)$, and their dimensions agree at every place, being determined by M alone:

$$\dim L_v = \begin{cases} \dim M^{G_v} & v \text{ finite} \\ v \nmid 2 \\ \dim M^{G_2} + 1 & v = 2 \\ \dim M^{G_\infty} - 1 & v = \infty. \end{cases}$$

So the conjecture would follow at once from

$$(\star) \quad L_v(E_a) = L_v(E_{-1/a}) \text{ for every place } v.$$

and $(\star)$ is automatic at all but finitely many places:

- **good v :** both conditions are the unramified subgroup $H_{\text{ur}}^1(\mathbb{Q}_v, M)$, which depends only on M ;
- **$v = \infty$:** $\Delta(y^2 = x^3 + k) = -432k^2 < 0$ for both curves, so both real loci are connected, $E(\mathbb{R})$ is divisible by 2, and $L_\infty = 0$ on both sides.

The bad places are the same for both curves — 2, 3 and the primes dividing a — so everything reduces to a finite check there.

4. The two descent maps differ by -1

There is one more piece of structure, and it says the two descents are not merely parallel but **aligned**.

Theorem B. Under the identification of Theorem A, the $\overline{\mathbb{Q}}$ -isomorphism between the two curves carries the descent map of E_a to λ times that of $E_{-1/a}$, where

$$\lambda = -1/\theta^2 = -(1/\theta)^2 \equiv -1 \quad \text{in } A_a^\times / (A_a^\times)^2.$$

Proof. $x^3 - a^{-1} = x^3 + au$ with $u = -a^{-2}$, so $E_{-1/a}$ is the sextic twist of E_a by u and the isomorphism over $\overline{\mathbb{Q}}$ is $(x, y) \mapsto (u^{1/3}x, u^{1/2}y)$. It carries roots to roots, so $\theta' = u^{1/3}\theta$, whence $u^{1/3} = \theta'/\theta = -1/\theta^2$; and then $X - \theta' = u^{1/3}(x - \theta)$. Finally $-1/\theta^2$ is -1 times a square. ■

Two readings. First, the quadratic part of the twist is by $u^{1/2} = a^{-1}\sqrt{-1}$: the involution $a \mapsto -1/a$ is **the quadratic twist by -1 , composed with a cubic twist** — and for $a = p^3$ the cubic part is trivial, so the original question is exactly about the quadratic twist by -1 . Second, χ_{-1} is ramified only at 2 and ∞ , which is why the two curves are so tightly coupled: at every odd place the twist is **unramified**.

5. Where the hypothesis “ a odd” lives

The conjecture is stated for odd a , and the computations say the restriction is not cosmetic. For odd a — positive and negative, prime, prime power and composite — the 2-Selmer ranks agree in every case tested. For even a they usually do **not**, and the failure is always by exactly one:

a	$\dim \text{Sel}_2(x^3 + a)$	$\dim \text{Sel}_2(x^3 - a^5)$	
2	1	0	differ by 1
6	0	1	differ by 1

a	$\dim \text{Sel}_2(x^3 + a)$	$\dim \text{Sel}_2(x^3 - a^5)$	
8	2	1	differ by 1
10	1	0	differ by 1
12	1	0	differ by 1
14	0	1	differ by 1
16	0	1	differ by 1
18	1	0	differ by 1

Fourteen of the sixteen even a tested disagree, every time by exactly 1. Since the local conditions agree at every good place and at ∞ regardless of parity, and since 2 is the only place where the quadratic part of the twist is ramified, this locates the whole content of the conjecture at $v = 2$: **oddness of a is what makes $L_2(E_a) = L_2(E_{-1/a})$** . A discrepancy of exactly 1 is what one expects from a single place where two maximal isotropic subspaces of a common $H^1(\mathbb{Q}_2, M)$ meet in a hyperplane.

6. Numerical verification

Sage’s `selmer_rank` is $\dim_{\mathbb{F}_2} \text{Sel}_2$; in PARI, `ellrank` returns $[r_1, r_2, s, L]$ with $C = T + r_2 + s$, where $T = \dim E(\mathbb{Q})[2]$ is 1 when $x^3 + a$ has a rational root and 0 otherwise.

```

dimtors2(E) = { my(f = factor(x^3 + E.a6));
  sum(i = 1, #f~, if(poldegree(f[i,1]) == 1, 1, 0)); }
C2(k) = { my(E = ellinit([0,0,0,0,k]), r);
  r = ellrank(E);
  if(dimtors2(E) > 0, 1, 0) + r[2] + r[3]; }

```

family	agree / total	comment
odd a , $ a \leq 25$	26 / 26	reproduces the original table
odd a , mixed prime powers and composites	19 / 19	3, 5, ..., 81, 99, 121, 125
odd $a < 0$	12 / 12	-3, ..., -49
even a	2 / 16	always off by exactly 1

7. Status

Proved here. Theorem A — the two curves have isomorphic mod-2 Galois modules, by the explicit algebra isomorphism $\theta \mapsto -1/\theta$. This is what makes the correlation structural rather than accidental, and it is what the question was asking for. Theorem B — the descent maps differ by -1 modulo squares, and the involution is the quadratic twist by -1 composed with a cubic twist, so that at every **odd** place the twist is unramified. The reduction of the conjecture to the local statement $(\star)$, together with its verification at the good places and at ∞ .

Not proved. The local statement $(\star)$ at $v = 2$, $v = 3$ and $v \mid a$. The even- a data says that $v = 2$ carries the real content and that oddness is exactly the hypothesis that makes it work; $v = 3$ and $v \mid a$ are odd places, where the quadratic part of the twist is unramified, so one expects them to be tractable by the standard local analysis — at an odd place of additive reduction, for instance, $E(\mathbb{Q}_v)/2$ is the 2-torsion of the component group, and the comparison is a finite computation with the Kodaira type.

A remark on what “equal” costs. Comparing two Selmer structures on a self-dual module, the Greenberg–Wiles formula gives only

$$\dim \text{Sel}(L) \equiv \dim \text{Sel}(L') + \sum_v \dim(L_v / (L_v \cap L'_v)) \pmod{2}.$$

So parity agreement already requires the local discrepancies to cancel modulo 2, and the observed **equality** is the stronger statement that they vanish one by one. That is why $(\star)$, and not a parity argument, is the right target.